Run report: LEAD-19 projected-base protocol, card 2, the pre-registered 20-repeat extension (2026-10-23T22:01Z)

Observations only (research-visuals skill, "Every experiment run"). The interpretation is in `../../README.md` the protocol is `../../PROTOCOL.md` card 1's report is `../run-20261023T2154Z/REPORT.md`.

What ran

item	value
why	PROTOCOL.md §9: on the five-repeat run the f2m15 published-rung interval (0.82, 1.21) straddled 1.10, so the same char2 command re-ran with `--repeats 20`, nothing else changed
card	`cards/CARD-lead19-projected-base-20261023-2.yaml` (sha256 `e7922074…5b04`), compiled by `experiment-director`, run by `executor-mechanical` through `python3 tools/task_card.py run`
receipt	`cards/CARD-lead19-projected-base-20261023-2.receipt.json` (sha256 `fb790e1e…1e1b`), status `completed`, two commands, one attempt each, exit 0
commit	`1cb4924a9aec75ea575969ce248619a558dabdd2` (the card's pin: card 1's records committed on top of the code commit `6783523a3`; the code is the same)
binary	`target/release/ic`, sha256 `373c0f1d…e2ec`, the binary card 1 bound; the build step reported nothing to do
command	`ic boundary --regime char2 --char2-degrees 15,18,21,24,27 --repeats 20 --seed 123212651130 --s4-max-degree 0 --base-arms raw,projected,subgroup --out research/lead19_projected_base/char2_repeats20.json`, 66.6 s
control	not re-run (the extension names the char2 command only); the analysis re-reads card 1's `prime24.json` for the control check, which passes unchanged
host	Intel Xeon @ 2.10 GHz (cloud container), 4 logical cores, linux x86_64; no isolation tier (counted operations)
raw record	`../../char2_repeats20.json` (5.9 MB); stdout and stderr under `../../receipts/`
analysis	`../../analysis_repeats20.json`, `../../analysis_repeats20.md`, by `docs/ic/tools/lead19_projected_base_compare.py` (the frozen plan, PROTOCOL.md §8); recorded values only

Arms, instances and seeds as in card 1; twenty targets per instance, of which repeats 0 to 4 are card 1's targets (same seeds, same counters) and repeats 0 to 2 the frozen round-5 targets. 5 instances × 20 repeats × 3 recorded arms (A, B, C; A′ is derived from A) × 2 rungs = 600 arm rows requested and completed; none failed or timed out.

Stop checks (PROTOCOL.md §6)

check	result
every row `verified`, `status: complete`	yes (`all_verified: true`)
prime-order control bit-identical across arms (card 1's record)	yes
arm A, repeats 0 to 2, equals the frozen round-5 counters on every char2 instance, both rungs	yes, no drift

Measured values

Same definitions as card 1's report (`S` in GAE/√r, every phase, cold; geometric mean of the per-repeat paired ratio with the 95 % percentile interval of 10,000 paired bootstrap resamples, seed 20261023; pooled yield).

row	h	rung	S(A)	S(A′)	S(B)	S(C)	S(A′)/S(B) (95 % CI)	S(A′)/S(C)	yield B/A	yield C/A	rho	S/rho: A′, B, C
f2m15	2	std (published)	27.75	15.88	12.35	12.16	1.29 (1.12, 1.52)	1.30	1.86	1.87	2.02	7.8×, 6.1×, 6.0×
f2m15	2	bal	24.02	24.02	18.20	17.98	1.31 (1.04, 1.69)	1.32	1.80	1.74	2.02	11.9×, 9.0×, 8.9×
f2m18	8	std	35.90	17.20	12.75	35.50	1.32 (1.20, 1.46)	0.47	3.78	3.69	1.91	9.0×, 6.7×, 18.6×
f2m18	8	bal	20.24	20.24	12.25	25.76	1.58 (1.30, 1.91)	0.73	3.67	3.41	1.91	10.6×, 6.4×, 13.5×
f2m21	2	std	23.35	15.46	10.11	10.83	1.51 (1.31, 1.73)	1.40	2.03	1.93	1.24	12.4×, 8.1×, 8.7×
f2m21	2	bal (published)	22.84	22.84	14.29	14.29	1.58 (1.35, 1.82)	1.59	1.96	1.90	1.24	18.4×, 11.5×, 11.5×
f2m24	8	std	34.66	24.02	17.08	27.79	1.39 (1.31, 1.48)	0.86	3.84	3.89	1.28	18.8×, 13.4×, 21.7×
f2m24	8	bal	22.75	22.75	10.94	16.49	2.01 (1.70, 2.34)	1.32	3.90	3.66	1.28	17.8×, 8.6×, 12.9×
f2m27	6	std	43.57	35.54	19.80	31.26	1.77 (1.57, 1.95)	1.12	5.78	5.90	1.02	34.8×, 19.4×, 30.6×
f2m27	6	bal (published)	50.99	50.99	14.75	18.84	3.36 (2.80, 4.02)	2.62	5.57	5.88	1.02	49.9×, 14.4×, 18.4×

Best rung per arm (by mean `S` over the twenty repeats) and the paired best-rung ratio: f2m15 std/std/std, S(A′)/S(B) 1.29 (1.12, 1.52); f2m21 std/std/std, 1.51 (1.31, 1.73); f2m27 A′ std, B bal, C bal, 2.41 (2.18, 2.67).

Per-repeat `S` on the three primary rows, published rung (twenty values, repeat order):

row	arm	S per repeat
f2m15 std	A′	13.0, 16.7, 13.6, 17.7, 17.9, 23.1, 16.5, 16.2, 16.3, 18.3, 9.7, 16.4, 12.1, 22.3, 18.7, 14.8, 11.5, 11.9, 12.5, 18.6

f2m15 std	B	18.2, 16.1, 15.1, 12.6, 17.0, 11.8, 9.0, 14.7, 9.2, 8.7, 10.4, 12.6, 12.1, 7.4, 15.6, 11.0, 11.3, 13.1, 11.6, 9.7
f2m15 std	C	12.5, 9.9, 12.4, 13.8, 14.0, 11.1, 10.1, 10.3, 10.8, 12.3, 9.8, 14.1, 12.2, 13.3, 18.7, 12.7, 11.5, 8.3, 9.6, 15.9
f2m21 bal	A′	16.9, 28.1, 28.2, 19.5, 22.1, 20.0, 24.0, 11.9, 17.9, 15.9, 31.6, 14.9, 17.8, 17.7, 33.6, 23.7, 26.0, 26.5, 35.7, 24.8
f2m21 bal	B	13.0, 16.1, 14.5, 14.5, 15.1, 18.3, 13.9, 17.2, 16.6, 8.5, 16.9, 15.7, 9.7, 10.7, 17.8, 12.4, 13.3, 7.5, 16.1, 18.0
f2m21 bal	C	14.1, 19.0, 9.2, 10.4, 9.0, 20.0, 13.1, 14.1, 14.7, 9.2, 18.8, 16.1, 21.0, 11.7, 11.8, 10.4, 13.9, 11.6, 20.9, 16.8
f2m27 bal	A′	25.5, 59.5, 25.7, 29.2, 67.4, 60.2, 42.7, 64.2, 60.2, 53.6, 68.3, 47.0, 51.7, 69.2, 44.4, 63.6, 31.4, 43.7, 71.0, 41.1
f2m27 bal	B	16.9, 7.5, 14.8, 13.1, 16.5, 19.5, 15.5, 18.3, 16.9, 12.5, 12.0, 12.9, 14.0, 12.9, 16.0, 16.9, 17.6, 14.4, 12.8, 14.0
f2m27 bal	C	21.9, 18.0, 16.5, 18.5, 13.8, 20.1, 20.6, 13.6, 21.9, 16.2, 18.7, 21.0, 18.9, 20.1, 26.2, 23.8, 12.5, 19.4, 15.0, 20.3

Trials and relations, primary rows, published rung, pooled over twenty repeats (sums of the per-repeat values in `analysis_repeats20.json`):

row	arm	trials (sum)	relations (sum)	relations per trial
f2m15 std	A	13,839	194	0.0140
f2m15 std	B	8,416	219	0.0260
f2m15 std	C	8,479	222	0.0262
f2m21 bal	A	403,731	372	0.00092
f2m21 bal	B	231,294	417	0.00180
f2m21 bal	C	231,445	406	0.00175
f2m27 bal	A	4,388,299	1,113	0.000254
f2m27 bal	B	1,001,477	1,415	0.00141
f2m27 bal	C	905,295	1,349	0.00149

`yield_over_ceiling_exact` (mean over repeats) on those rows: A 1.06 / 1.02 / 1.03, B 1.08 / 1.00 / 0.98, C 0.99 / 1.00 / 1.06 (f2m15 / f2m21 / f2m27): every arm realises its exact counting ceiling; the projection raises the ceiling, not the realisation.

Phase split of `S` at the std rung, mean over the twenty repeats, read from the record's `factor_base` and `relations` phases (the linear algebra and verification phases are the small remainder):

row	h	A base / relations	B base / relations	C base / relations
f2m15	2	13.76 / 13.84	2.01 / 10.19	1.91 / 10.10
f2m18	8	24.56 / 11.23	6.38 / 6.24	29.18 / 6.20
f2m21	2	11.59 / 11.73	3.76 / 6.32	3.71 / 7.09
f2m24	8	23.69 / 10.94	13.34 / 3.71	24.08 / 3.68
f2m27	6	22.72 / 20.84	14.86 / 4.93	26.51 / 4.74

Class census, base-phase counters and the class test removed for A′ are the same as in card 1's report (same curves, same bases).

Pricing band: arm C's trace test at w = 0, 0.1, 1 GAE moves S(C) by at most 0.25 (f2m15 std, 12.16 ⟶ 12.41); every A′ > B and A′ > C order holds across the band.

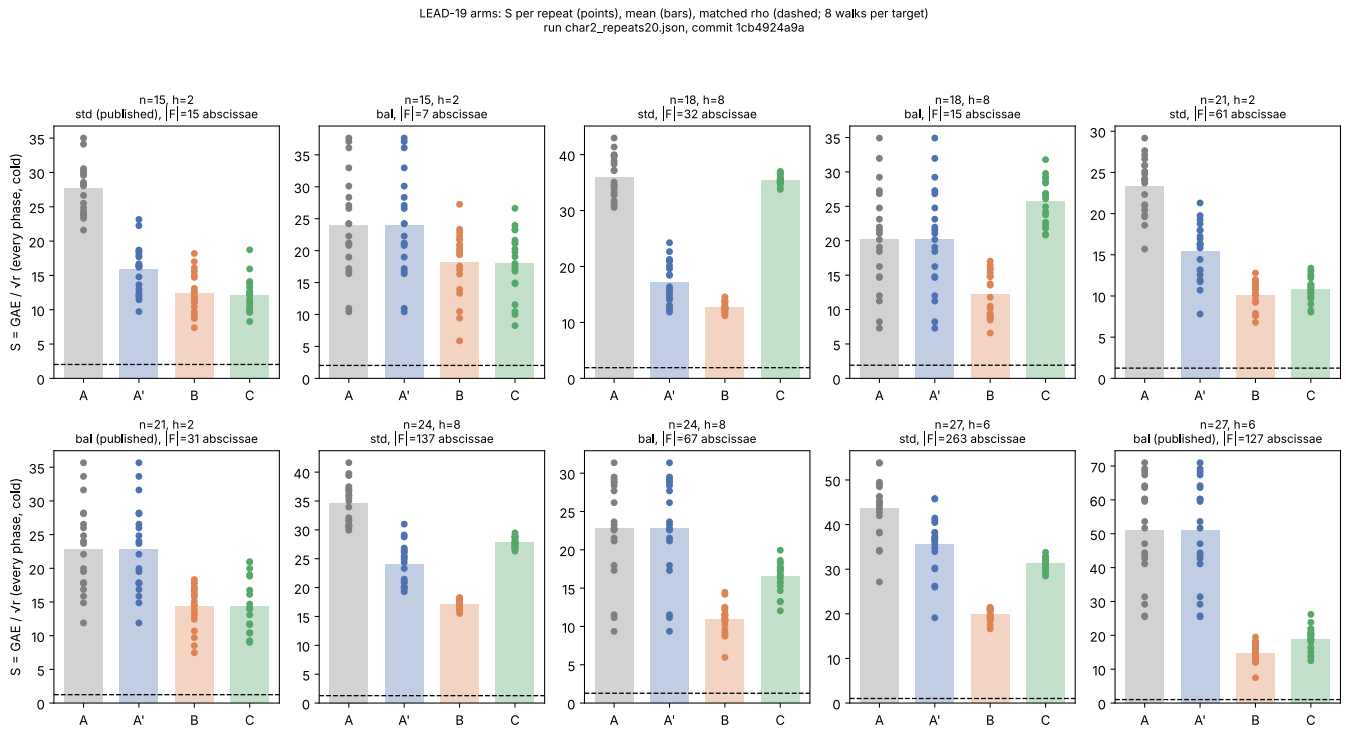
Predictions (PROTOCOL.md §7), as the analysis script evaluated them:

row	P1 (predicted)	P2 yield B/A (band)	P3 S(A′)/S(B) published rung (predicted)	P4 yield B/C	P5
f2m15	0.502 (0.50): within 10 %	1.86 ([1.4, 2.6]): holds	1.29, CI (1.12, 1.52) (1.4–1.5)	0.99: within 15 %	identical
f2m21	0.501 (0.50): within 10 %	1.96 ([1.4, 2.6]): holds	1.58, CI (1.35, 1.82) (1.4–1.5)	1.03: within 15 %	identical
f2m27	0.167 (0.17): within 10 %	5.57 ([4.2, 7.8]): holds	3.36, CI (2.80, 4.02) (about 3)	not applicable	identical

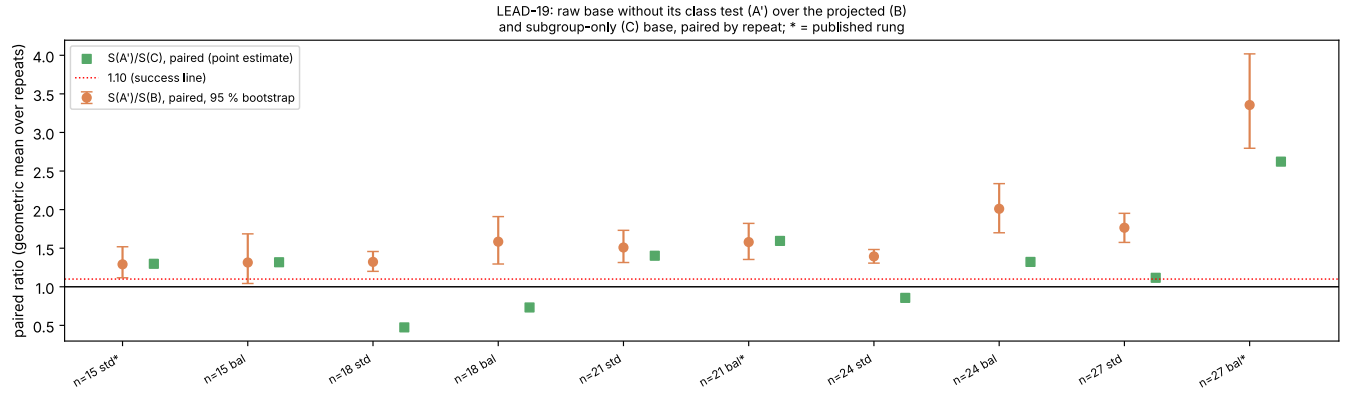
Published-row ratio (accounting, not the lever; frozen round-5 mean S over this run's mean): f2m15 26.31 / 12.35 = 2.13 (B), 2.16 (C); f2m21 24.39 / 14.29 = 1.71 (B), 1.71 (C); f2m27 36.93 / 14.75 = 2.50 (B), 1.96 (C).

The frozen comparison's verdict line (PROTOCOL.md §9): **success on rows n = 15, n = 21, n = 27**.

Graphs



S per arm, rung and repeat (twenty repeats); dashed line the matched rho



Paired ratios S(A′)/S(B) with the bootstrap interval and S(A′)/S(C), against the 1.10 line

Sources beside the graphs: `../../plot_lead19.py` reading `../../analysis_repeats20.json`.

Missing or not measured

- As in card 1: no isolation tier, no wall-clock claim, no S_4 arm, and no replay certificate, validator report or `VALIDATION.json` yet.
- The record's `host.git_commit` is `null` (the binary was built outside a git-aware step); the card's receipt pins the commit.